

# ALLAN ILYASOV

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## SUMMARY

Driven software engineer with hands-on experience building and deploying Gen/Agentic AI-powered applications and assisting a consumer business with tech enablement, analytics, and better decision-making. Motivated by difficult problems where data, engineering, and real-world impact intersect; skilled in machine learning and novel AI, full-stack development, and cloud infrastructure; experience designing consumer products with an eagerness to continuously learn in this rapidly evolving field.

## EXPERIENCE

### *Manager – Uncle Edik’s Pickles*

**December 2019 – Present**

- Grew a home-based pickle start-up and supported its growth into a brick-and-mortar factory and national brand.
- Trained & led a team of 5 employees in production of thousands of jars per week of 9 different pickled products.
- Engineered Apps Script automations in Google Sheets to streamline wholesale invoicing and business analytics.
- Managed sales to customers, wholesale delivery to supermarkets, and inventory replenishment from Restaurant Depot.
- Designed branded merchandise, and produced viral marketing content, generating millions of views.
- Ensure compliance with NYS Food Safety regulations & maintained quality control with daily production checklists.

### *Graduate Research Assistant – St. John’s University*

**September 2025 – May 2026**

#### **Codify AI – Advancing Education Through Generative AI & LLM (Dr. Fazel Keshtkar):**

- Product manager & lead architect of [codifyai.org](https://codifyai.org), an AI-powered programming tutor using LangChain agents, Claude Haiku 4.5 via AWS Bedrock, and RAG for streaming responses, and context-aware tutoring in Java, Python & SQL.
- Designed production-ready AWS infrastructure (Elastic Beanstalk, DynamoDB, Route 53, HTTPS) with Docker containerization, supporting scalable REST API architecture.
- Built React-based adaptive learning platform with persistent state management, real-time progress tracking.
- Earned “Commercializable Project” award at the 2025 St. John’s University’s Student Research Conference.
- Published research in the FLAIRS-39 (2026) proceedings. [doi.org/10.32473/flairs.39.1.141554](https://doi.org/10.32473/flairs.39.1.141554)

## SELECT PROJECTS

### **TaxSnap – Receipt Management App** | [github.com/allanilya/TaxSnap](https://github.com/allanilya/TaxSnap)

- Built an iOS app letting users scan receipts offline, auto-extracting merchant, date, and amounts with on-device OCR.
- Integrated OpenAI embeddings with KNN classification for automatic receipt categorization across 100+ merchants.
- Implemented Core Data + CloudKit sync and multi-format export (CSV, Excel, PDF) for IRS-compliant tax records.

### **Time Series Forecasting for Financial Markets** | [github.com/allanilya/Stock-analytics](https://github.com/allanilya/Stock-analytics)

- Created ARIMA, LSTM, & GRU forecasting models across AAPL, NVDA, & LYFT, reducing RMSE by up to 82%.
- Implemented grid search across 36 ARIMA configurations and built 2-layer LSTM/GRU architectures with dropout regularization utilizing Python, TensorFlow, and Statsmodels.

### **Diabetes Prediction Model** | [github.com/allanilya/Diabetes-Prediction](https://github.com/allanilya/Diabetes-Prediction).

- Trained Random Forest, Logistic Regression & KNN models with 5-fold cross-validation to predict diabetes risk.
- Investigated lifestyle and demographic predictors of diabetes using 230K+ patient records.
- Prepared data through imputation, encoding, outlier detection and addressed class imbalance leveraging SMOTE.

## EDUCATION

### **St. John’s University – Queens, New York**

#### *Master of Science, Data Science*

**May 2026**

#### *Bachelor of Science, Computer Science*

**May 2025**

- Dean’s List 2021-2025 | Member of the Association for Computing Machinery | EPIC Mentoring Program, Student Mentor

## SKILLS SUMMARY

- **Programming Languages:** Python, Java, SQL, R, JavaScript, Swift, TypeScript, C++, PHP
- **Frameworks & Libraries:** React, Next.js, Django, Flask, Spring Boot, Node.js, Bootstrap, LangChain, LangGraph, REST APIs, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch, Spark, Beautiful Soup, NLTK, OpenCV
- **Cloud & DevOps:** AWS, Docker, DynamoDB, Elastic Beanstalk, Lambda, API Gateway
- **Linguistic Skills:** Russian native speaker | Intermediate proficiency in Spanish

**Work Eligibility:** US Citizen